

USDA/EGRET model

30 September 2026

Overall question: How does temperature affect germination?

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- ▶ What are the average overall responses to chilling and forcing?
- ▶ And how does it vary by species? (And what is λ ?)
- ▶ And dormancy classes?

How the model works

Grab from provenance model?

$$\hat{y} = \alpha_0 + \alpha_{provenance} + \beta(C)$$

$$\alpha_{provenance} \sim MVN(0, \sigma_{\alpha}^2)$$

$$y \sim normal(\hat{y}, \sigma_y^2)$$

Results so far ... forcing/chilling

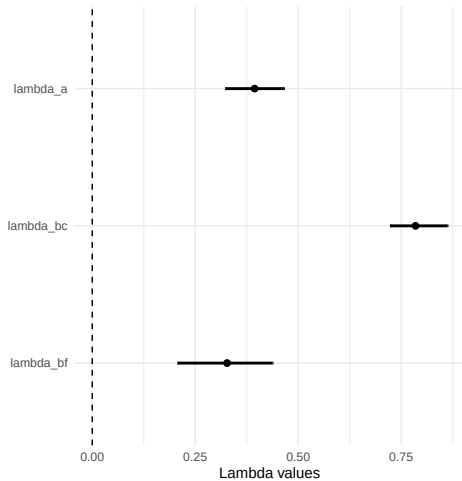
Overall: Gymnosperms vs. angiosperms

Results so far ... forcing/chilling and intercept

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Results so far ... forcing/chilling

Phylogenetic effects: intercept, chilling, forcing (angio and gymno)



Results so far ... forcing/chilling

Many species!

Results so far ... forcing/chilling

Dormancy classes

Other things we should look at?

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- ▶ Seed size?